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Dynamic Friction-aware Lagrangian Network for Accurate GRF Estimation in Legged Robot

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Abstract: Accurate Ground Reaction Force (GRF) estimation is crucial for legged robots to perform complex tasks, yet conventional sensor-based methods suffer from fragility and impracticality. Indirect estimation is challenged by intricate system dynamics and the complex, dynamic nature of friction. This paper proposes a novel MIMO Hysteretic Friction-aware Lagrangian-Based Network (M-Net) for sensorless GRF estimation. M-Net is comprised of a Lagrangian-based Network (L-Net) for modeling energy-conservative rigid-body dynamics and a Hysteresis-aware Network (H-Net) utilizing Temporal Convolutional Networks (TCNs) to capture dynamic friction characteristics, including hysteresis and coupling effects. By integrating physics-informed constraints into the loss function, enhanced data efficiency and model generality are achieved. The performance of M-Net is validated through MuJoCo simulations on a quadruped robot leg, where it is compared against conventional DeLaN and classical friction compensation methods. Results demonstrate that M-Net significantly improves joint torque and GRF estimation accuracy, enabling robust sensorless GRF prediction by precisely modeling the complex dynamics of systems with dynamic friction.

Keywords: Lagrangian Mechanics, Time-series Neural Network, Ground Reaction Force, Dynamic Friction

1. INTRODUCTION

As robotic applications become increasingly diverse and complex, research on legged robots has gained significant interest. Unlike wheeled platforms, legged robots utilize their limbs to navigate intricate and unstructured environments, offering immense potential for various missions [1]. During these operations, legged robots continually interact with their surroundings through ground contact. Achieving precise and robust locomotion necessitates the accurate estimation of contact forces between the robot's feet and the environment, commonly known as Ground Reaction Force (GRF) [2].

Accurate GRF estimation is essential for implementing advanced control strategies, such as impedance control, by endowing robot legs with compliance. This capability enables legged robots to execute dynamic maneuvers, including jumping and landing [3], [4]. Thus, robust GRF perception is fundamental for realizing the diverse movement capabilities required by legged robots.

Traditionally, GRF has been directly measured using force sensors mounted on the robot's feet [5]. However, this approach presents limitations such as sensor susceptibility to impact damage and impracticality due to the need for numerous sensors [6]. To address these challenges, research has shifted towards indirect methods for estimating external forces without direct sensors [7], [8]. A common drawback, however, lies in their reliance on overly simplistic leg models.

Real-world robotic systems are inherently non-conservative, encompassing energy-conservative rigid-body dynamics and energy-dissipating elements like friction. Accurately modeling these dissipative components is crucial for precise system identification and control. Existing research often simplifies leg dynamics and crudely approximates nonlinear elements like friction. Traditional mathematical models struggle with unknown parameters and limited complexity, while Neural Networks (NNs) are highly data-dependent [9]. Physics-informed Neural Networks (PINNs) [10], such as Deep Lagrangian Networks (DeLaN) [9], improve generalization by embedding physical principles. However, most DeLaN-based methods considering friction typically employ classical friction models, neglecting dynamic characteristics [11].

A significant challenge arises from the dynamic nature of friction, exhibiting complex phenomena like stick-slip and hysteresis influenced by past conditions [12]. Friction can also demonstrate position-dependent behavior [13]. To capture these nonlinear and dynamic phenomena, Time-series Neural Networks (TNNs) are suitable. While Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) have been used [14], they suffer from limited long-term memory and inefficient parallel computation. To address this, the Temporal Convolutional Network (TCN) has been proposed [15]. TCN is well-suited for time-series data and leverages causal convolution to effectively model dynamic characteristics, making it superior to other TNNs for long-term dependencies.

In this paper, a novel approach is proposed that leverages TCNs to effectively model the system's dynamic friction characteristics, integrated with a DeLaN-based method for energy-conservative system dynamics. The performance of this combined method is evaluated through simulations involving a quadruped robot leg and GRF estimation.

The contributions of this paper are summarized as follows:

- A novel method is proposed that combines a Lagrangian-based network and a time-series network to effectively learn the dynamic characteristics of both energy-conservative systems and friction that causes energy dissipation.
- To demonstrate the effectiveness of the proposed

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method, the leg of a quadruped robot is modeled, and GRF is estimated in a simulation environment.

The remainder of this paper is organized as follows: Section 2 analyzes the system to identify and model friction occurring in real systems. Section 3 describes each component of the proposed method and presents the structure for GRF estimation. Section 4 presents the simulation results for system modeling and external force estimation to demonstrate the effectiveness of the proposed method. Finally, Section 5 concludes the paper and discusses future works.

2. DYNAMIC PROPERTIES OF FRICTION

Friction, an inherent element causing energy dissipation, is inevitably present in real mechanical systems. However, its uncertain nature makes it challenging to distinguish from the system's inherent dynamics. Friction is generally predominantly influenced by velocity, leading to many proposed models in the form of velocity-dependent functions. To accurately identify such uncertain friction, various models have been proposed [12].

2.1 DYNAMIC FRICTION MODEL

Representative models that can effectively represent dynamic characteristics include Dahl's model and the LuGre model. Both equations are expressed as dynamics that consider the deformation of bristles (or contacts) on the system surface. Since both models are effective in representing dynamic friction characteristics like stick-slip, they are used in friction models to precisely control systems like linear actuators [16].

Dahl's model (F_{dahl}) is mathematically expressed as follows:

$$\frac{dF_{dahl}}{dx} = \sigma \left(1 - \frac{F_{dahl}}{F_c}\,\mathrm{sgn}(v)\right)^{\alpha}, \qquad (1)$$

where σ is the stiffness coefficient and α is a parameter that determines the shape of the stress-strain curve. F_c is the Coulomb friction force, and F_{dahl} cannot exceed this value. x represents the relative displacement of the contact interface, and v is the relative velocity between the two surfaces. This model considers not only the static characteristics of friction but also the influence of past states (velocity). However, compared to static friction models, their coefficients are less intuitive, requiring more effort for identification.

2.2 TIME-SERIES NEURAL NETWORK

To replace the equation-based identification techniques that require significant effort, attempts have been made to model friction using NNs. However, typical NNs merely create a complex continuous function and cannot effectively reflect the dynamic characteristics of friction that change over time. To overcome the static nature of NNs, research has progressed on modeling dynamic characteristics using TNNs. In particular, the Temporal Convolutional Network (TCN) method has most effectively modeled phenomena such as hysteresis [15].

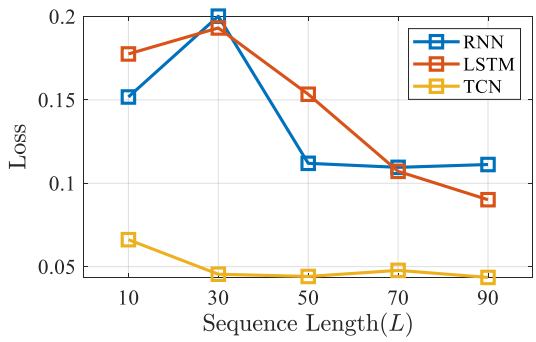

Fig. 1. Learning loss comparison for each TNNs case

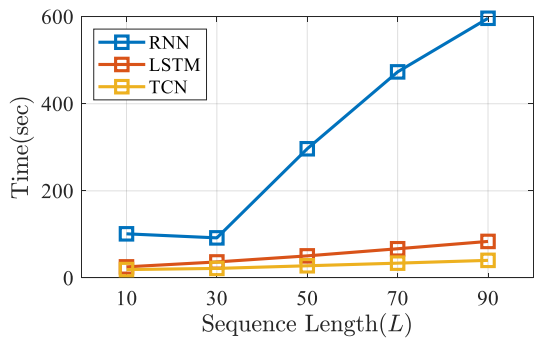

Fig. 2. Learning time comparison for each TNNs case

To experimentally verify the dynamic characteristic modeling capabilities of TNNs, model (1) is modeled using several TNN techniques. The parameters of the model used for testing are $\sigma = 50$, $F_c = 1.5$, and $\alpha = 1$, respectively. Three types of networks are used for training: RNN, LSTM, and TCN. Since each technique processes time-series data, they all share a common parameter called sequence length (L), which determines the size of the batch used for training. Sequence length is a key parameter affecting modeling performance, and we compared the training results by varying this value.

Fig. 1 shows that TCN's dynamic characteristic modeling performance is superior for all values of L. At $L = 30$, where the difference is largest, TCN's loss decreased by approximately 76% compared to LSTM, and at $L = 90$, where the difference is smallest, a difference of about 52% occurred. While RNN and LSTM show significant changes in training performance depending on L, TCN is relatively robust to this variable.

Furthermore, Fig. 2 shows the training time for each case. Unlike RNN, where training time increases exponentially as L increases, LSTM and TCN show a relatively linear increase. In particular, TCN's structure allows for efficient parallel computation, resulting in shorter training times and better model performance. These results confirm that TCN is effective for learning models that include dynamic characteristics without effort for identification. Therefore, this paper aims to effectively learn the dynamic characteristics of friction using TCN.

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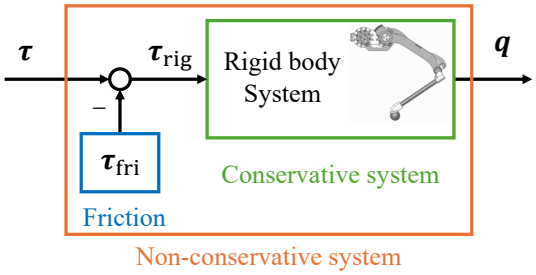

Fig. 3. Scheme of energy non-conservative system

3. MIMO HYSTERETIC FRICTION-AWARE LAGRANGIAN-BASED NETWORK

Real-world systems are inherently non-conservative. They can be conceptualized as comprising an energy-conservative rigid-body system and friction components that cause energy dissipation, as illustrated in Fig. 3. Since the conservative rigid-body system adheres to Lagrangian mechanics, it is modeled by a Lagrangian-based Network (L-Net). Conversely, because friction, which causes energy dissipation, exhibits dynamic characteristics, it is modeled using a time-series network, specifically a Hysteresis-aware Network (H-Net).

3.1 LAGRANGIAN-BASED NETWORK: L-NET

The Lagrangian-based Network is proposed to model energy-conservative systems [9]. In this approach, the Multi-Layer Perceptron (MLP) structure of conventional NNs is combined with Lagrangian mechanics, leveraging the strengths of both to enhance data efficiency by complementing model complexity and generalization capabilities. The rigid-body dynamics derived from Lagrangian mechanics are expressed as:

$$\mathbb{M}_{\text{rig}}(\boldsymbol{q})\,\ddot{\boldsymbol{q}} + \mathbb{C}_{\text{rig}}(\boldsymbol{q},\dot{\boldsymbol{q}}) + \mathbb{G}_{\text{rig}}(\boldsymbol{q}) = \boldsymbol{\tau}_{\text{rig}}, \tag{2}$$

where $\mathbb{M}_{\text{rig}} \in \mathbb{R}^{n\times n}$ is the mass matrix, and $\mathbb{C}_{\text{rig}}, \mathbb{G}_{\text{rig}} \in \mathbb{R}^{n}$ are the Coriolis/centrifugal and gravitational forces, respectively, for an n-DoF rigid body system. Here, the position-dependent $\mathbb{M}_{\text{rig}}$ and $\mathbb{G}_{\text{rig}}$ are estimated from an MLP, with $\mathbb{C}_{\text{rig}}$ then estimated through additional matrix operations:

$$\mathbb{C}_{\text{rig}}(\boldsymbol{q},\dot{\boldsymbol{q}}) = \dot{\mathbb{M}}_{\text{rig}}(\boldsymbol{q})\dot{\boldsymbol{q}} - \frac{1}{2}\left(\frac{\partial}{\partial \boldsymbol{q}}\left(\dot{\boldsymbol{q}}^{\top}\mathbb{M}(\boldsymbol{q})\dot{\boldsymbol{q}}\right)\right)^{\top}. \tag{3}$$

In essence, the energy-conservative rigid-body dynamics are effectively learned by the L-Net.

3.2 HYSTERESIS-AWARE NETWORK: H-NET

To effectively learn dynamic friction characteristics, TCN is employed. Causal convolution operations, which accurately capture time-series data characteristics, are performed by the TCN. This effectiveness is particularly observed in learning friction that exhibits dynamic properties, such as hysteresis. Given that friction forces in real systems can depend on both velocity and position, sequences of position and velocity are used as inputs to the

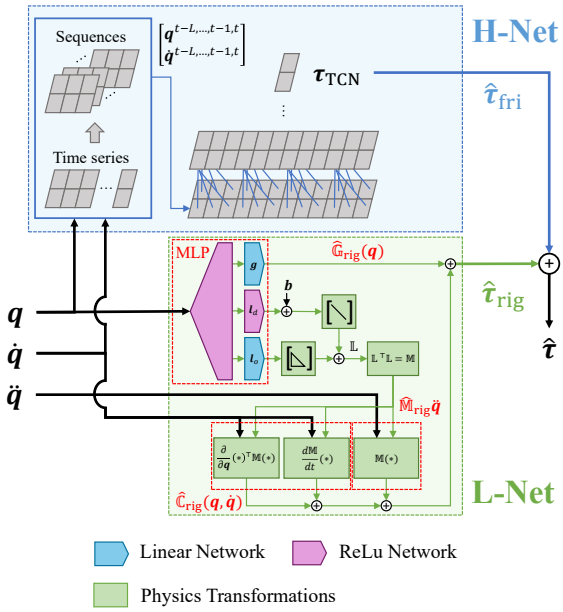

Fig. 4. Scheme of Mysteric-Net (M-Net) combining L-Net with H-Net.

TCN for friction estimation:

$$\hat{\boldsymbol{\tau}}_{\text{fri}} = \boldsymbol{\tau}_{\text{TCN}}\left(\boldsymbol{q}^{(t-L,\ldots,t-1,t)}, \dot{\boldsymbol{q}}^{(t-L,\ldots,t-1,t)}\right). \tag{4}$$

In multi-degree-of-freedom systems, variations in load and system characteristics occur with the system's pose, leading to coupling effects where friction forces between joints are influenced by each other [17], [18]. Through the input of all joint positions and velocities into the TCN, these coupling effects are accounted for by the friction model. This network is referred to as H-Net.

3.3 MYSTERIOUS NETWORK: M-NET

To model non-conservative systems, a combined network of L-Net and H-Net is proposed, which is termed the MIMO Hysteretic Friction-aware Lagrangian-based Network, or simply Mysteric Network (M-Net). The structure of M-Net is depicted in Fig. 4. The output of M-Net is the sum of rigid-body dynamics and friction forces, formulated as:

$$\begin{aligned}\hat{\boldsymbol{\tau}} &= f^{-1}(\boldsymbol{q},\dot{\boldsymbol{q}},\ddot{\boldsymbol{q}}) \\ &= \hat{\mathbb{M}}_{\text{rig}}(\boldsymbol{q})\,\ddot{\boldsymbol{q}} + \hat{\mathbb{C}}_{\text{rig}}(\boldsymbol{q},\dot{\boldsymbol{q}}) + \hat{\mathbb{G}}_{\text{rig}}(\boldsymbol{q}) + \hat{\boldsymbol{\tau}}_{\text{fri}}.\end{aligned} \tag{5}$$

Since M-Net incorporates a Lagrangian structure, its training can be guided not only by force estimation but also by a loss function based on energy change. Unlike conventional NNs, enhanced model generality and increased data efficiency are achieved through the inclusion of physics-informed constraints in the loss function. Specifically, the energy change, considering energy dissipated by friction, is given by:

$$\frac{d\mathbb{E}_{\text{rig}}}{dt} = \frac{d\mathbb{T}_{\text{rig}}}{dt} + \frac{d\mathbb{V}_{\text{rig}}}{dt} = (\boldsymbol{\tau} - \boldsymbol{\tau}_{fri})\dot{\boldsymbol{q}}, \tag{6}$$

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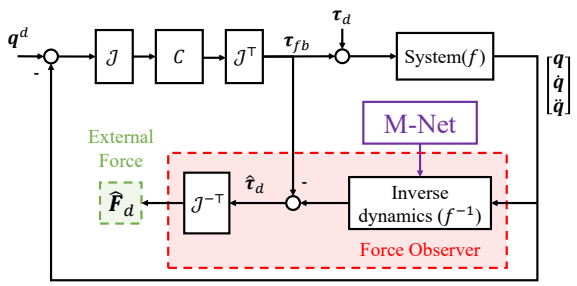

Fig. 5. Scheme of FOB using M-Net based inverse dynamics.

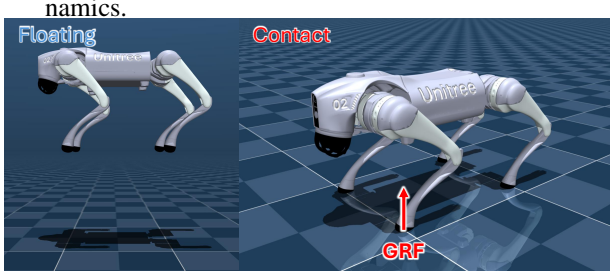

Fig. 6. MuJoCo simulation environment. Unitree's Go2 model is used. Training dataset is acquired in floating case and GRF data is acquired in contact case.

where

$$\frac{d\mathbb{T}_{\text{rig}}}{dt} = \frac{1}{2}\dot{\boldsymbol{q}}^\top \mathbb{M}_{\text{rig}}(\boldsymbol{q})\dot{\boldsymbol{q}},$$
$$\frac{d\mathbb{V}_{\text{rig}}}{dt} = \frac{\partial \mathbb{V}_{\text{rig}}}{\partial \boldsymbol{q}}\frac{\partial \boldsymbol{q}}{\partial t} = \mathbb{G}_{\text{rig}}(\boldsymbol{q})\dot{\boldsymbol{q}}.$$

where $\mathbb{T}_{\text{rig}}$, $\mathbb{V}_{\text{rig}}$, and $\mathbb{E}_{\text{rig}}$ are kinematic, potential and total energy, repectively. The second term of (6) consists only of the position-dependent mass matrix and gravitational force, which can be estimated from the L-Net's MLP structure. This means that not only inverse dynamics but also results derived from energy change can be used to train the M-Net model. The loss function used for M-Net training is comprised of two components: a force loss (l_τ) and an energy loss (l_E).

$$l_{total} = l_\tau + l_E, \tag{7}$$

where

$$l_\tau(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}, \boldsymbol{\tau}) = \sum_{i=1}^{N} \left|\hat{\boldsymbol{\tau}}_i - \boldsymbol{\tau}_i\right|^2,$$
$$l_E(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}, \boldsymbol{\tau}) = \sum_{i=1}^{N} \left|\frac{\widehat{d\mathbb{E}}_{\text{rig},i}}{dt} - (\boldsymbol{\tau}_i - \hat{\boldsymbol{\tau}}_{\text{fri},i})^\top \dot{\boldsymbol{q}}_i\right|^2.$$

N represents the number of data samples used for training. M-Net is trained to minimize (7), ensuring that both rigid-body dynamics and the energy dissipated due to friction are incorporated into the learning process. Consequently, M-Net is effective for modeling system dynamics, including energy dissipation.

3.4 FORCE OBSERVER

By accurately modeling the inverse dynamics of a quadruped robot leg system using M-Net, forces in the task space (Cartesian coordinates) can be estimated via a Force Observer (FOB) structure [7]. The FOB structure, shown in Fig. 5. $\boldsymbol{q}^d$ represents the desired position, $\mathcal{J}$ is the Jacobian matrix mapping from joint space to Cartesian space, $\mathcal{C}$ is the feedback controller, $\boldsymbol{\tau}_{fb}$ is the feedback control input and $\boldsymbol{\tau}_d$ denotes joint disturbances resulting from external forces. These external forces ($\boldsymbol{F}_d$), which generate the joint disturbances in joint space, are the primary target for estimation by the FOB.

$$\hat{\boldsymbol{F}}_d = \mathcal{J}^{-\top}\left(f^{-1}(\boldsymbol{q}, \dot{\boldsymbol{q}}, \ddot{\boldsymbol{q}}) - \boldsymbol{\tau}_{fb}\right). \tag{8}$$

Kinematic information of the multi-degree-of-freedom system is required for the Jacobian ($\mathcal{J}$) included in this structure. In the following section, simulations will be presented demonstrating how the quadruped robot leg model can be learned using M-Net, and how, in conjunction with the FOB, GRF applied in the Cartesian coordinate system can be estimated.

4. SIMULATION

In this paper, the performance of M-Net in learning the dynamics of a quadruped robot leg system with dynamic friction is validated by comparing its modeling accuracy against other methods. Furthermore, the influence of inverse dynamics on GRF estimation performance is investigated by employing a FOB to estimate GRF. The simulations are performed using the MuJoCo physics engine (Google DeepMind) with the Unitree Go2 model from the MuJoCo Menagerie, as illustrated in Fig. 6.

Each leg is assumed to have a biarticular structure (m : hip joint, b : knee joint) and is operated in SLIP coordinates. A simple P-controller is used for feedback control of leg movements. To account for position-dependent dynamic friction characteristics and coupling effects between joints, the following friction model($\boldsymbol{\tau}_{\text{dyn}}$) is applied to the leg system:

$$\frac{d\boldsymbol{\tau}_{\text{dyn}}}{dt} = \sigma\left(1 - \frac{\boldsymbol{\tau}_{\text{dyn}}}{F_c}\,\text{sgn}(\boldsymbol{v})\right)^\alpha \boldsymbol{v}, \tag{9a}$$

$$\tau_{\text{dyn,m}}[k+1] = \tau_{\text{dyn,m}}[k] + \frac{d\tau_{\text{dyn,m}}}{dt}\Delta t + \beta_m q_m + \gamma_m q_b, \tag{9b}$$

$$\tau_{\text{dyn,b}}[k+1] = \tau_{\text{dyn,b}}[k] + \frac{d\tau_{\text{dyn,b}}}{dt}\Delta t + \beta_b q_b + \gamma_b q_m, \tag{9c}$$

where $\boldsymbol{\tau}_{\text{dyn}} = [\tau_{\text{dyn,m}} \quad \tau_{\text{dyn,b}}]^\top$ is the friction torque vector for hip and knee joint, k is the discrete time index, the sampling time is 2ms, and the parameters are set as $\sigma = 10$, $F_c = \alpha = 1$, $\beta_m = \beta_b = 0.003$, and $\gamma_m = \gamma_b = 0.001$. For learning the leg system that incorporates this friction model, a total of three methods are employed:

- C1: Conventional DeLaN (rigid-body dynamics only, no friction modeling).
- C2: L-Net + classical friction model (Coulomb, viscous, and Stribeck friction compensation).

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Table 1. Hyperparameters for learning and networks

Learning Parameter		Network Parameter	
Learning rate	5.0×10^{-4}	MLP in L-Net	# of hidden layers: 2
Weight initialization	Xavier Normal		# of nodes: 16
Batch size	1000	H-Net	# of hidden channels: 8
Weight decay	1.0×10^{-5}		Sequence length(L): 30
Diagonal epsilon	1.0×10^{-2}		Kernel size: 3

Table 2. Loss comparison for each case

	C1	C2	C3
l_{total}	30.507	1.962	0.043

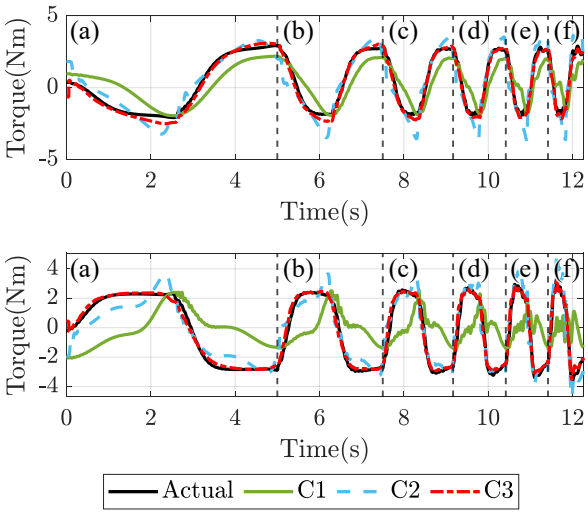

Fig. 7. The comparison of torque estimation for each case. (Top) hip joint, (Bottom) knee joint

- C3 (Proposed method): L-Net + H-Net (hysteresis and coupling effects across joints).

For C2, the coefficients of the classical friction model are learned alongside the L-Net's MLP.

Table 1 lists the heuristically determined learning and network parameters used for each training case, with the learning parameters and the MLP structure within the L-Net remaining consistent across all cases.

4.1 MODEL ACCURACY

The extrapolation performance of M-Net is evaluated through torque estimation. For training, data is acquired from squatting motions performed by the robot in a floating state (no external forces, as shown in the left part of Fig. 6). This ensured that only friction and rigid-body dynamics are captured. The squatting motion involved the retraction and extension of the legs based on a sinusoidal reference, with all four legs following the same reference. To verify the model's generality, training data is collected from motions at frequencies of 0.1, 0.3, ..., 0.9 Hz, while testing data is collected from motions at frequencies of 0.2, 0.4, ..., 1.2 Hz.

Table 2 shows the converged loss values for each method after training with data acquired in the floating case. Similar initial losses (around 10) are observed for all methods. However, the greatest reduction in the final converged loss is clearly achieved by C3, indicating its superior ability to accurately capture system characteristics, including dynamic friction.

Table 3. 2 Norm error of torque estimation

$\|\|e\|\|_2/\sqrt{N}$	C1	C2	C3
q_m	1.34×10^{-2}	9.6×10^{-3}	3.1×10^{-3}
q_b	2.86×10^{-2}	9.8×10^{-3}	3.4×10^{-3}

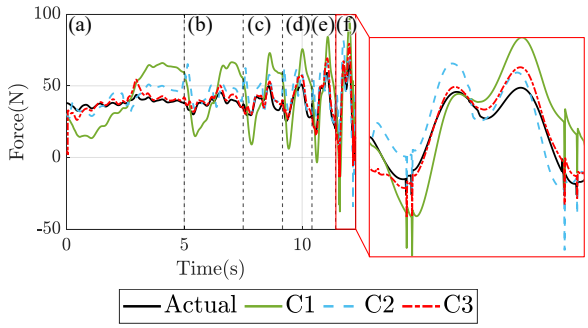

Fig. 8. The comparison of GRF($F_{d,z}$) estimation for each case.

Table 4. 2 Norm error of GRF

$\|\|e\|\|_2/\sqrt{N}$	C1	C2	C3
F_z	0.2346	0.1148	0.0613

The torque estimation results for frequencies (labeled (a) to (f) respectively) are presented in Fig. 7. As expected, very poor estimation performance is observed for C1, as it does not account for friction. Even with the inclusion of static friction characteristics, worse performance is still exhibited by C2 compared to C3, as C3 effectively models both dynamic friction and coupling effects.

Table 3 compares the 2-norm errors for torque estimation across each joint. A similar trend to the loss comparison is observed, with C3 exhibiting the highest torque estimation accuracy, followed by C2 and C1. This result highlights the varying data utilization efficiency and model structure effectiveness.

4.2 GRF ESTIMATION

The learned models from the previous subsection can be integrated into the FOB structure to estimate external GRF. Unlike the torque estimation simulation, GRF data is acquired during robot-ground contact. Similar to the previous simulation, data is collected for on-the-spot squatting motions at frequencies of 0.2, 0.4, ..., 1.2 Hz while the robot is in contact with the ground. The GRF in the vertical direction($F_{d,z}$) is estimated during robot movement, with actual GRF values obtained through MuJoCo's internal sensor functionalities.

The estimated GRF for various frequencies is illustrated in Fig. 8. Actual GRF values are obtained from ground sensors, while the estimated values are derived using the models of each case in conjunction with the FOB structure. Overall, C3 demonstrated superior performance in estimating actual GRF, a finding quantitatively supported by Table 4. This confirms that the accuracy of the leg model directly affects the external force estimation performance. Consequently, precise learning of rigid-body dynamics, including friction in multi-degree-of-freedom systems, enables accurate GRF estimation

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through the FOB structure.

5. CONCLUSION

A method for estimating GRF without direct sensors is presented in this paper, by training a leg system using two distinct network architectures. To effectively model the dynamic friction characteristics inherent in real robot leg systems, a time-series network, H-Net, is employed. The remaining rigid-body dynamics are learned using an L-Net. These two networks are combined to form M-Net. To validate its performance, joint torque estimation and Cartesian GRF estimation experiments are conducted within a MuJoCo simulation environment. It is demonstrated by our findings that by tailoring network architectures to specific system characteristics, both data efficiency and model accuracy can be significantly improved, ultimately enabling accurate GRF estimation without the need for physical sensors.

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